



Scientific visualisation & computational geometry

INSAIgo training session - Louis Gombert 29/04/26



Agenda for today

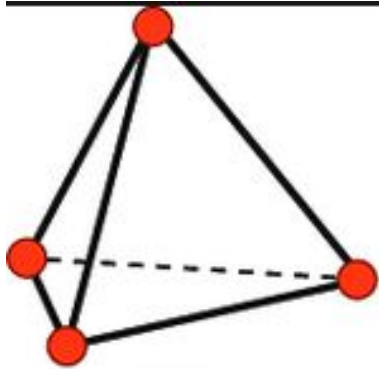
- 3D geometry basics
- Scientific visualisation basics
- Practice: isosurfacing
- Extra exercises



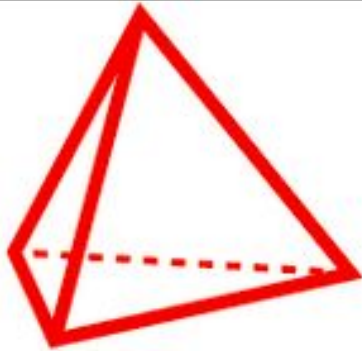
\$ whoami

- Louis Gombert
- TC2023
- Ex INSAIgo, Cod'INSA, CVA, SIA...
- R&D Engineer @ Kitware
- OSS contributor & activist

3D Geometry basics



Vertex
(sommets)



Edge
(bord)



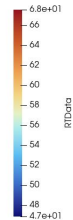
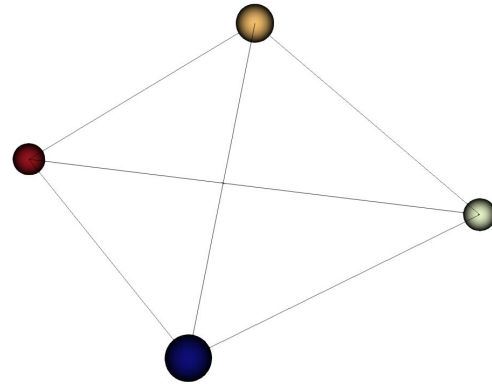
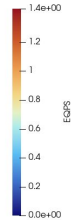
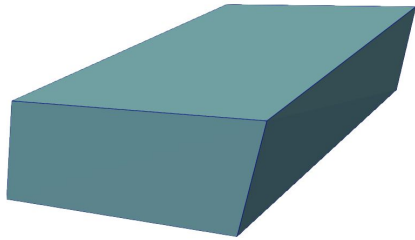
Face



Cell
(cellule)

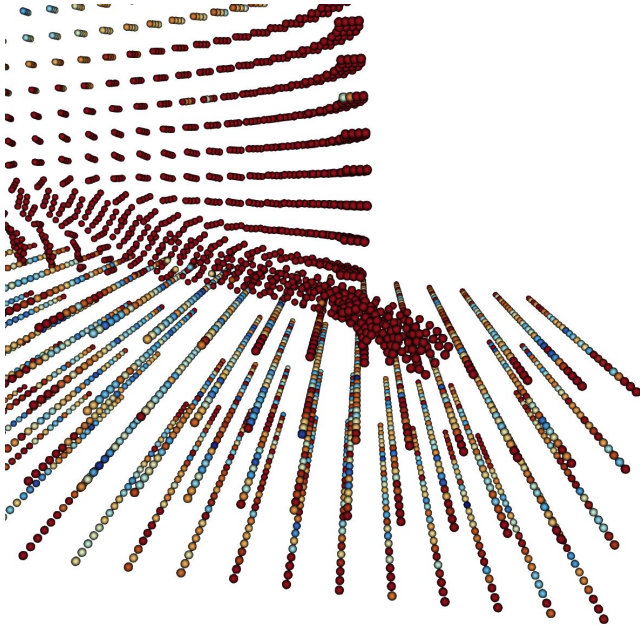


Storing information in the geometry





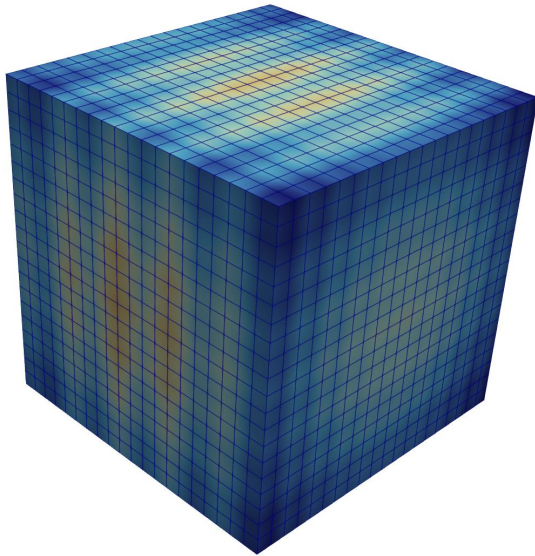
Mesh types: point clouds



- Coordinate list

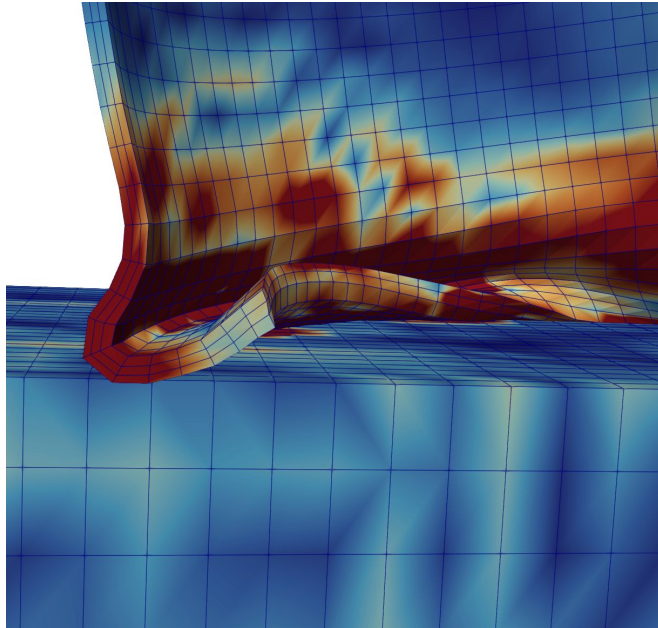


Mesh types: regular grids



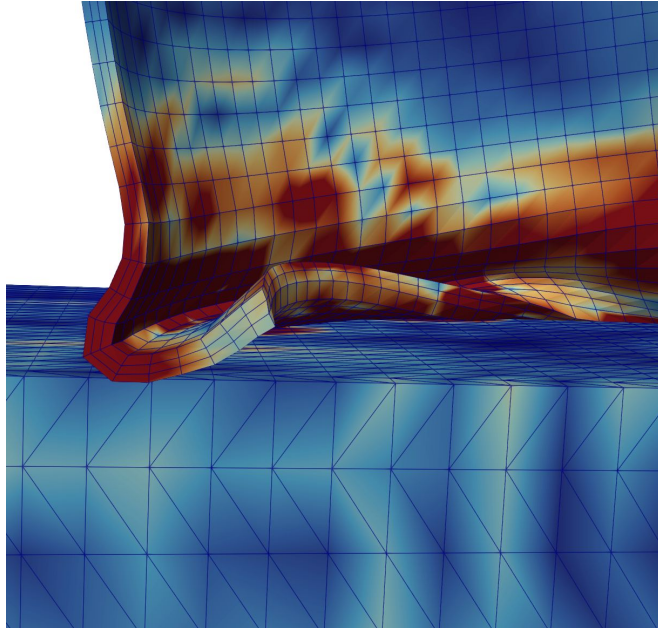
- Dimensions
- Origin
- Spacing

Mesh types: unstructured grids (1 cell type)



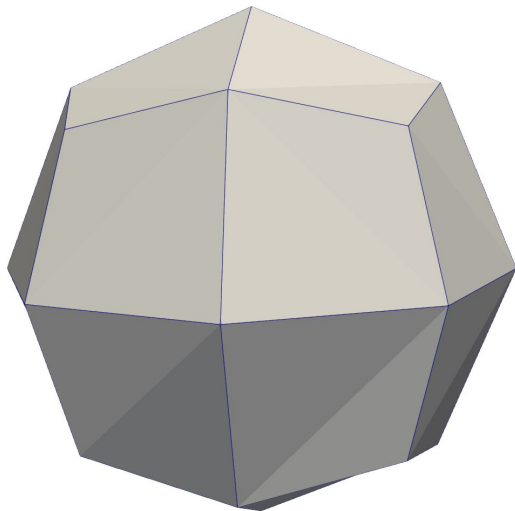
- Cell Type (1)
- Point positions
- Connectivity (cell $\leftrightarrow$ point mapping)

Mesh types: unstructured grids (multiple types)



- Cell Types
- Point position
- Connectivity (cell $\leftrightarrow$ point mapping)
- Offsets (if cell type has a dynamic # of points)

Mesh types: generalized polyhedrons



- Cell Types
- Point position
- Connectivity (cell $\leftrightarrow$ point mapping)
- Offsets (if cell type has a dynamic # of points)
- Face connectivity (face $\leftrightarrow$ point mapping)
- Face offsets

Eye candy



Credits: J. Favre @ CSCS, the birth of Jupiter (2024)



Scientific visualization: challenges

- How to represent 3D data on a 2D screen ?
 - Slices
 - Isocontours
 - Streamlines
 - Volume rendering
 - Many more...

Isosurfaces

Surfaces where the variable attached to vertices is constant

= level set

Applications : [CFD](#), medical imaging ([CT](#)), astrophysics...

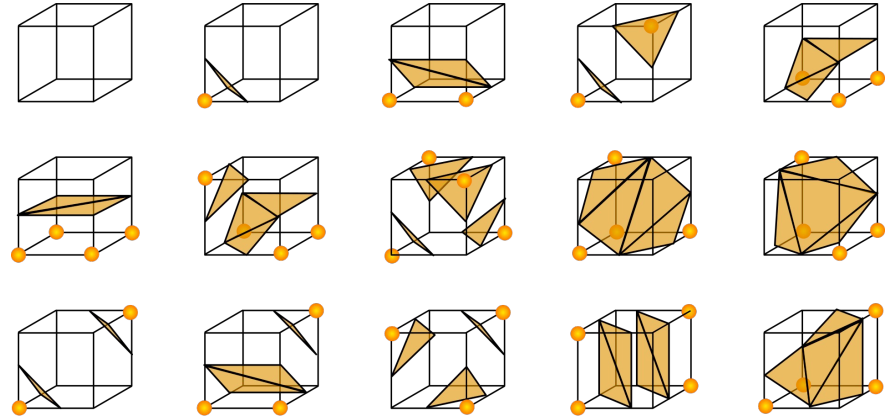
Usually we only use the 2-manifold, closed surface (contour)





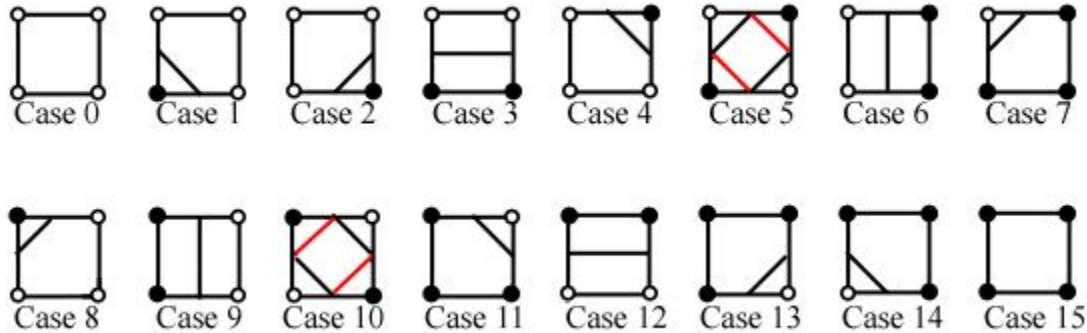
Marching cubes algorithm

Published in 1987 (patented) and improved over time



- Before : pre-calculate all 256 possibilities
- For each cube in the mesh :
 - Calculate which points are over the threshold
 - Generate the right triangles from pre-computed data
 - Interpolate coordinates
- Assemble the isosurface from the triangles
- Trim to only keep the closed surface

Implementation time ! (in 2D)



Fill all the TODOs : <https://gist.github.com/Lgt2x/6395df29cff113a782f374a4abc502a4>

Complete version : <https://gist.github.com/Lgt2x/14ef956d5af2d5429b065332bc213401>



Improving the algorithm...

- Remove duplicate points
 - Support triangular cells
 - Support general polygons
 - Support 3D cells
-
- What if we have a point cloud?
 - Can it be faster?